

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461211.7125 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.011
Transit depth (Rp/Rs)^2	2.63 +/- 0.35 [%]
Orbital Inclination (inc)	89.95 +/- 0.52
Airmass coefficient 1 (a1)	0.9239 +/- 0.0086
Airmass coefficient 2 (a2)	0.052 +/- 0.035
Scatter in the residuals of the lightcurve fit is	0.93 %
Best Comparison Star	#3 - [513, 303]
Optimal Aperture	3.81
Optimal Annulus	15.2
Transit Duration (day)	0.0961 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.162 ± 0.011 vs 0.14762 (deviation 1.31σ)
Duration fit vs published	0.0961 d vs 0.1033 d (deviation 3.62σ)
Mid-time O-C	-1.2 min
Depth SNR	14.7
Residual scatter	0.93 %

Light curve

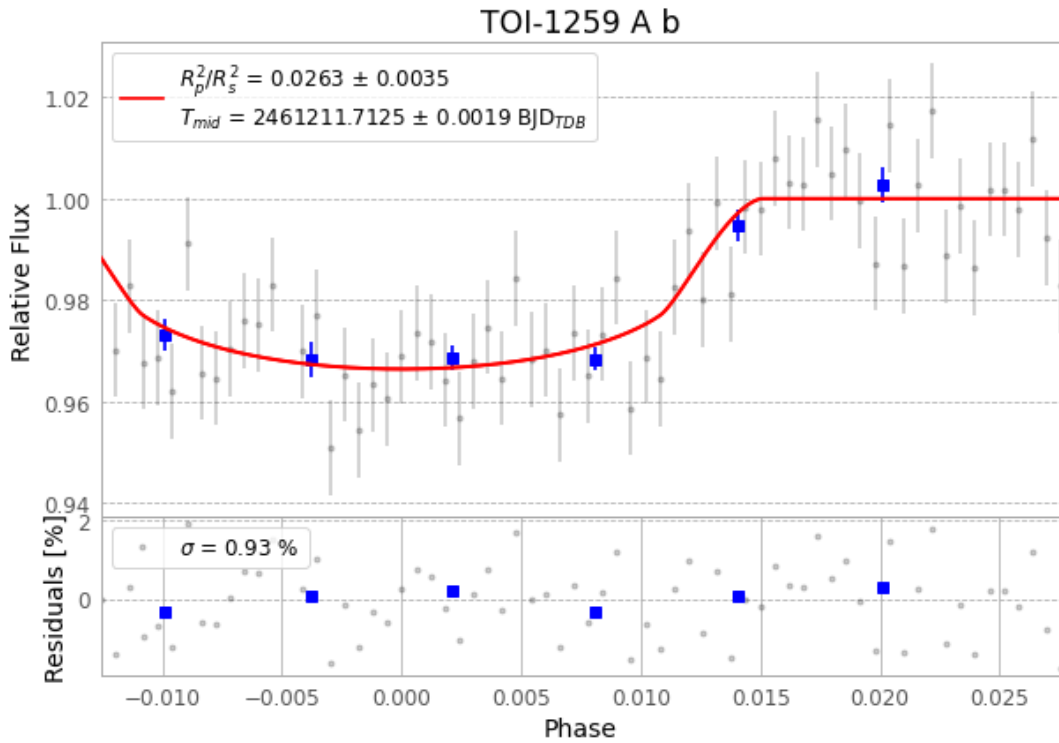


Figure 1 — FinalLightCurve_TOI-1259 A b_2026-06-19.png (SHA-256 8237d6e0d9971197...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 296], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4b20469e693aca22...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

68 FITS frames (45 MB), TOI1259260620040315.FITS → TOI1259260620072415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1259-260620.log (8fcc36c90a4d...), FinalLightCurve_TOI-1259 A b_2026-06-19.png (8237d6e0d997...), FinalLightCurve_TOI-1259 A b_2026-06-19.csv (ab2a6fde3ce3...)

Compute resources

Wall time 107.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 23:58Z.